DIGITAL TWIN SYSTEM ARCHITECTURE DIAGRAM

Happiest Health Summit 2026 — End-to-End Technical Flow

Layer	Components	Function / Role
Data Sources	Synthetic Indian EHR + Wearable IoT Generator	500 patients, CGM (5m), HR/HRV (1m), Sleep, Steps
Data Ingestion	EHRLoader, SensorLoader, StreamSimulator	Schema validation, time-series alignment, real-time generator
Processing Store	EHRPreprocessor, SensorPreprocessor, Fusion	Sliding window generation (72 steps), HRV extraction, normalization
Digital Twin Core	BiLSTM+Attention, XGBoost, Multi-Modal NN	2-hour glucose forecasting, ASCVD risk scoring, joint representations
Clinical Actions	PatientDigitalTwin, AlertSystem, What-If	Hysteresis debouncing, 4-tier alert triage, counterfactual simulation
Doctor Dashboard	Streamlit Clinician Portal (Port 8501)	Live telemetry, risk gauges, SHAP attribution, scenario simulator

Key Mathematical & Architectural Formulations:

• **BiLSTM + Self-Attention:** Context vector $c = \sum \alpha_t h_t$ where $\alpha_t = \text{softmax}(w^T \tanh(W h_t + b))$, decoding 2-hour future glycemic concentration.

• **TreeSHAP Attribution:** Explains non-linear feature importance for individual patient cardiovascular hazard calculations.

• **Gated Multi-Modal Fusion:** Dynamic gate $g = \sigma(W [h_{\text{temporal}}, h_{\text{static}}] + b)$ dynamically weighting time-series vs longitudinal clinical priors.

• **Debouncing Alert Logic:** Prevents alarm fatigue with minimum hysteresis cooldown window $\Delta t_{\text{cooldown}} = 30$ mins.